



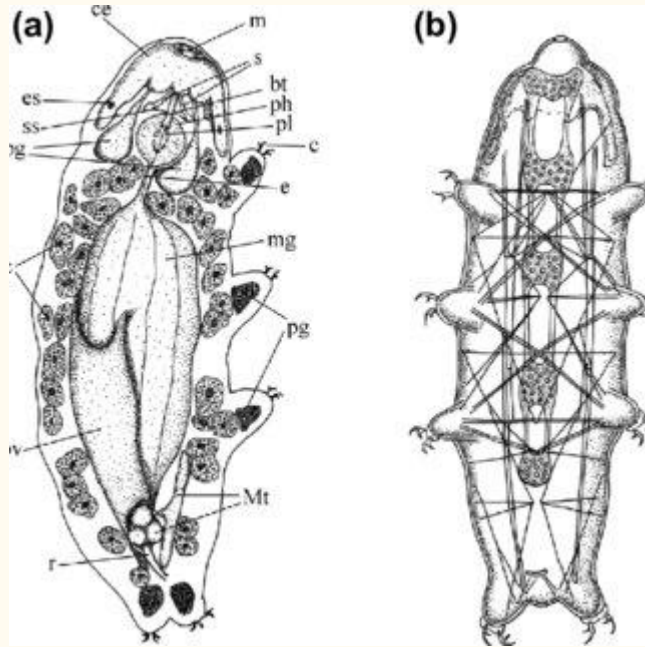
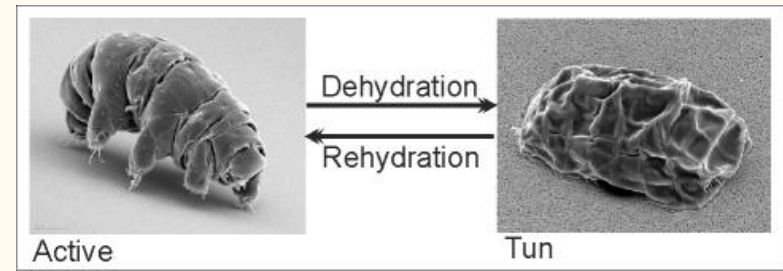
Tardigrades

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Physics 135

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Ryan Erickson
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Tardigrades (Water Bears)



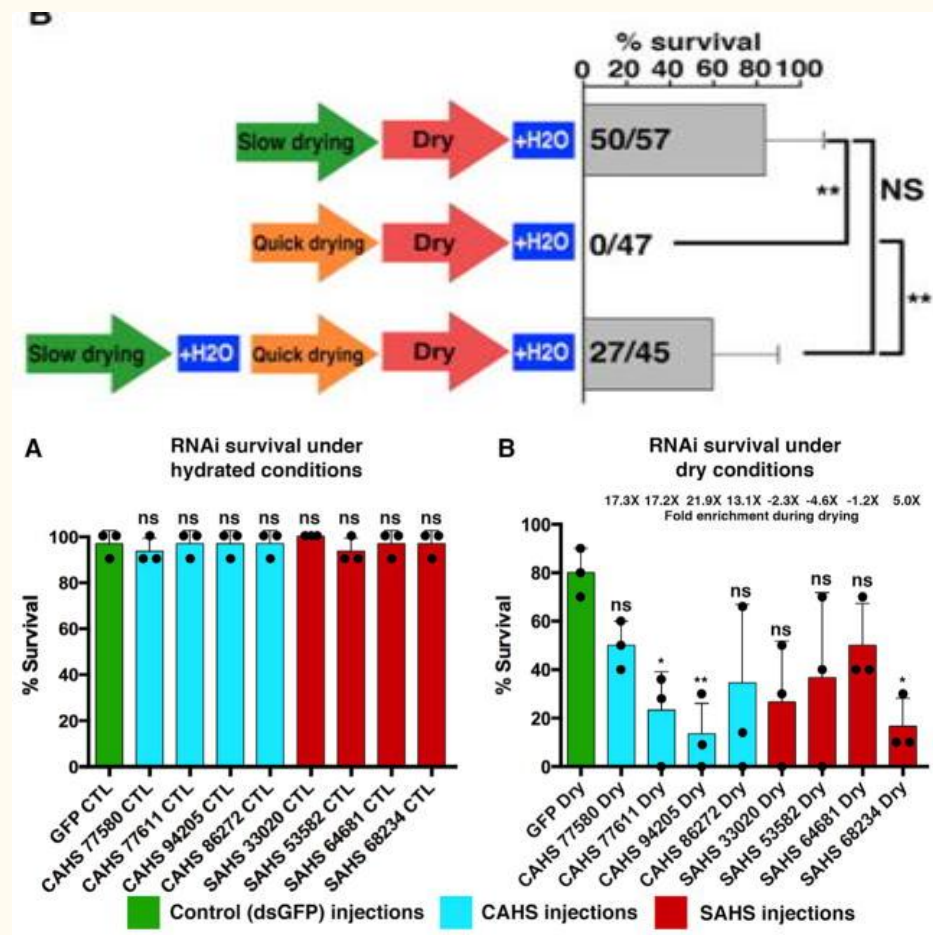
Desiccation

- The Tun State
 - Metabolism $< .01\%$
 - Water content $\sim 1\%$
- Trehalose
 - Disaccharide
 - Very low levels
- Why?
 - IDPs



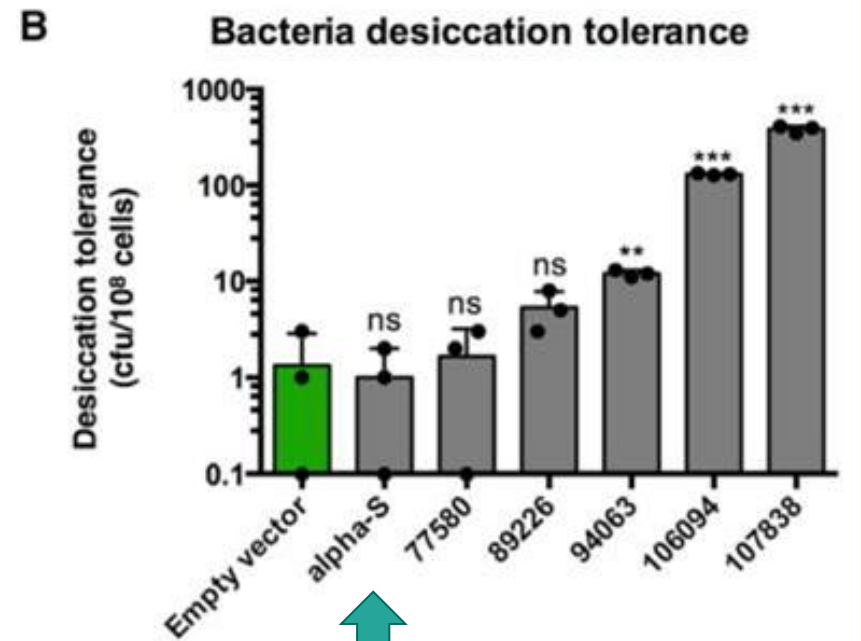
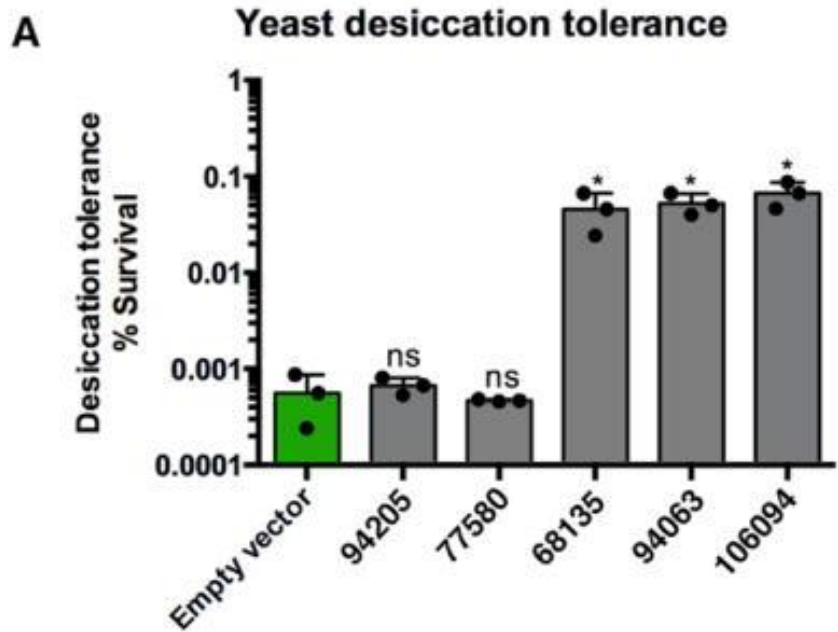
Anhydrobiosis and TDPs

- Preconditioned vs sudden
 - Protectant?
- Protein families thought to encode IDPs
 - CAHS (Cytosolic)
 - SAHS (Secreted)
- TDPs
- IDPs ~ stress resistance
- Stress → TDP production
- RNA interference (RNAi)



Boothby, et. al. Tardigrades Use Intrinsically Disordered Proteins To Survive Desiccation 2018

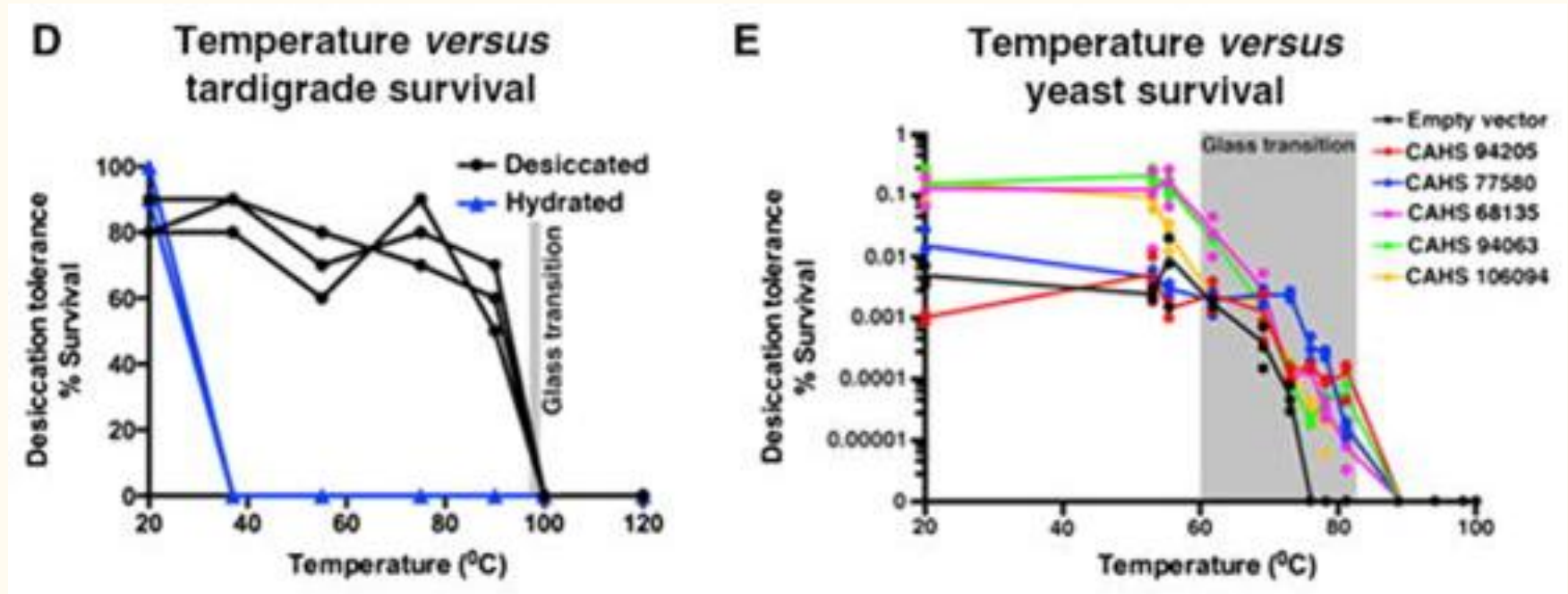
TDPs in other Organisms



Boothby, et. al. Tardigrades Use Intrinsically Disordered Proteins To Survive Desiccation 2018

Common IDP, but no increase in desiccation tolerance

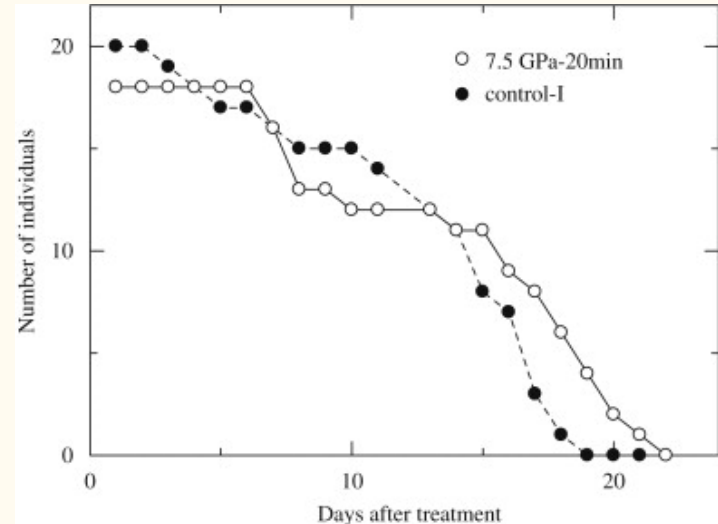
The Mechanism (vitrification)



Boothby, et. al. Tardigrades Use Intrinsically Disordered Proteins To Survive Desiccation 2018

Extreme Pressures

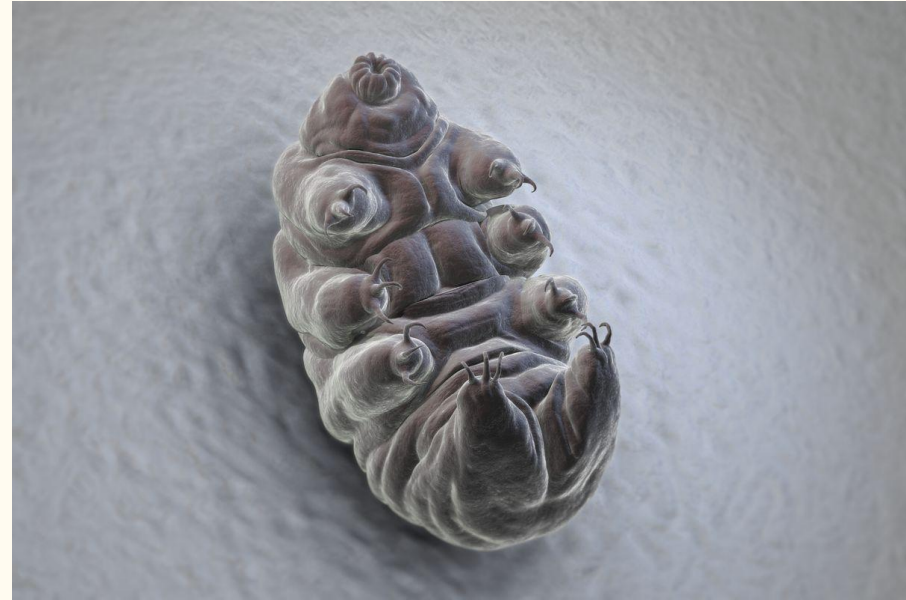
- Up to 13 hours of 7.5 GPa pressure
 - Equivalent to pressure at 180km below Earth's Surface
- Up to 10 days of extremely low pressure (vacuum)



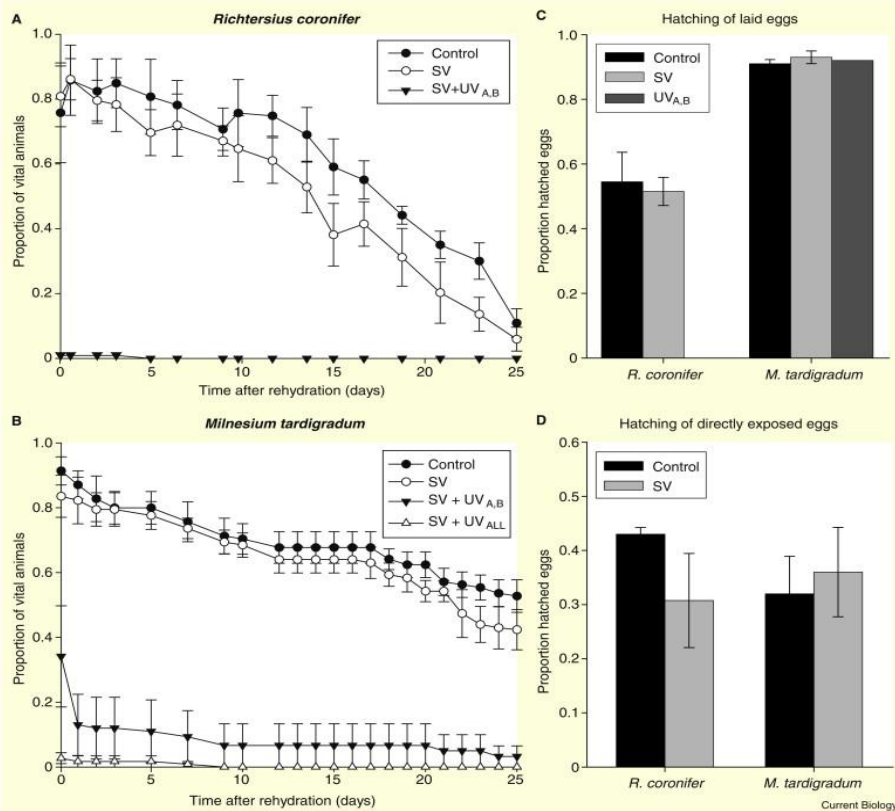
Ono, et. al. Effect of high hydrostatic pressure on to life of the tiny animal tardigrade (2008)

Extreme Temperatures

- Below -273°C and above 90°C in Tun state
- Below -196°C and above 120°C for short periods of time (<30 min) in normal state
- Possibility to survive near-zero temperatures during interplanetary travel



Tardigrades survive space vacuum and radiation



Experiment:

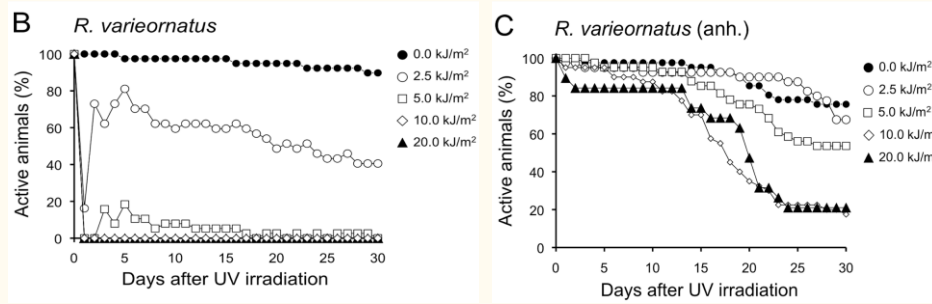
- Orbit radius: 258-281 km
- Period of experiment: 10 days

Conclusions:

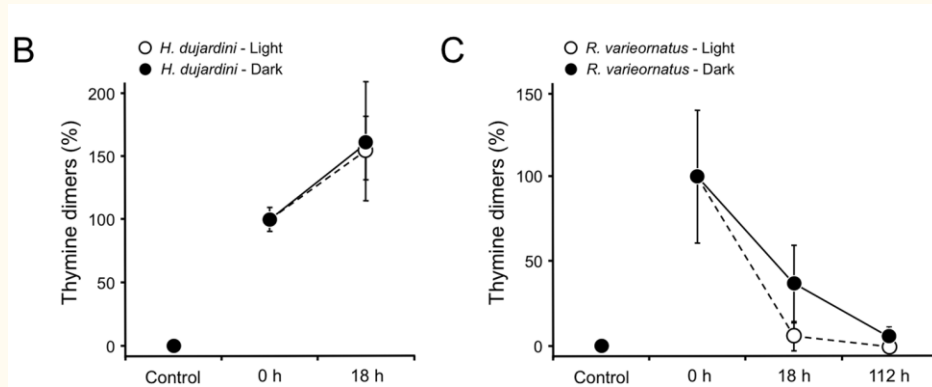
- Tardigrades survived in vacuum
- *M. Tardigradum* survives vacuum + radiation
- Mechanism of survival remains a mystery!

Jonsson et al. Tardigrades survive exposure to space in low Earth orbit (2008)

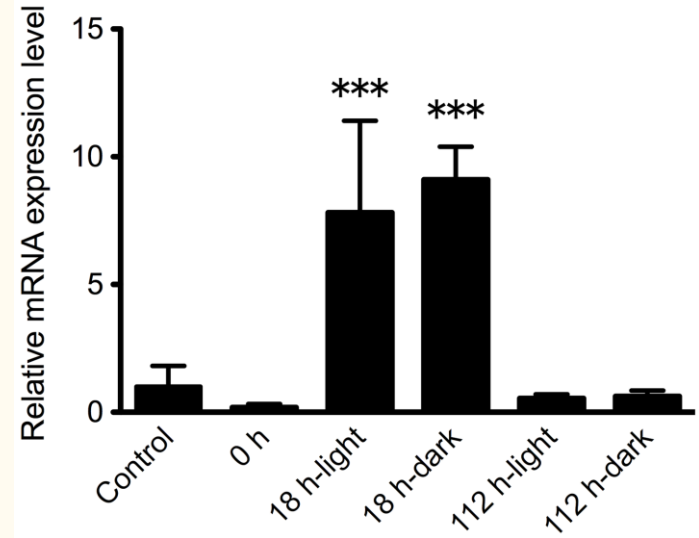
DNA repair and protection in Tardigrades



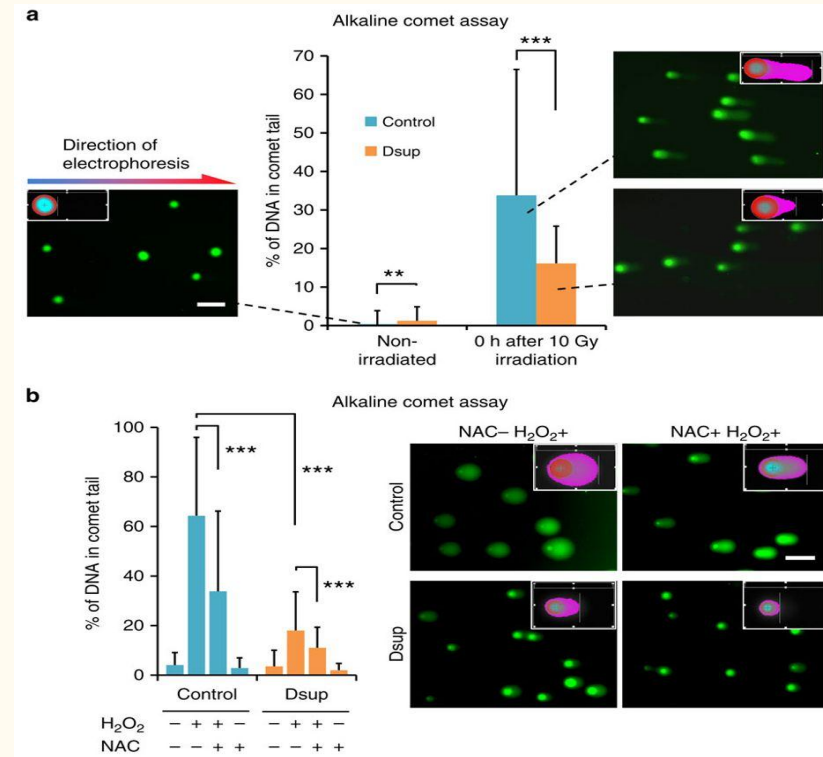
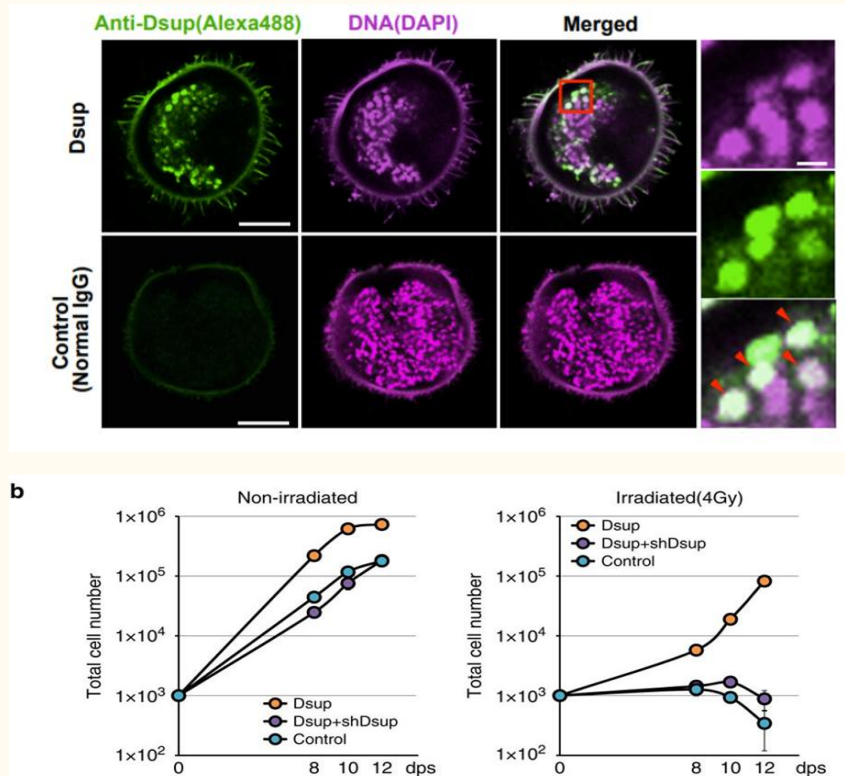
- Photolyases gene “phrA” responsible for DNA repair network
- CPD DNA Photolyase breaks thymine dimers



Horikawa et al. *Analysis of DNA repair and Protection in the Tardigrade Ramazzottius varieornatus and Hypsibius dujardini after exposure to UVC Radiation*. PLOS communication. 2013



Damage Suppressor Protein is a novel mechanism for survival



Hashimoto et al. *Extremotolerant tardigrade genome and improved radiotolerance of human cultured cells by tardigrade-unique protein.* Nature Communications. 2016

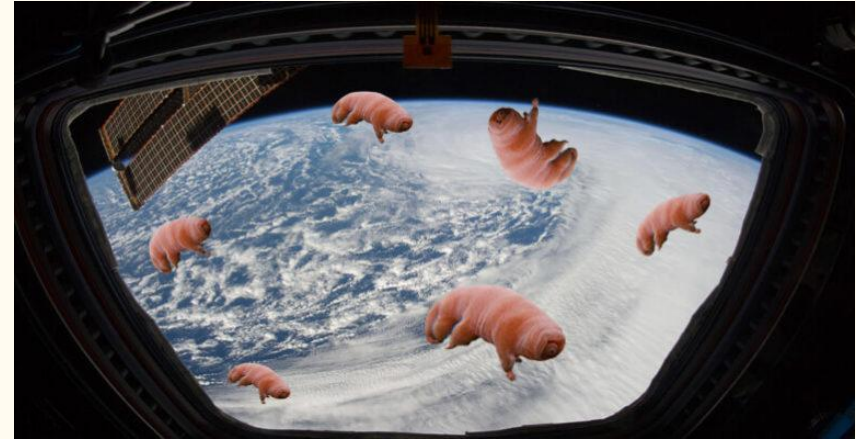
Summary

- Desiccation - using CAHS and SAHS
 - These dry-tolerant proteins are intrinsically disordered in water but develop secondary structures in the dehydrated state that allow them to stabilize DNA, proteins, and cell membranes
- Can survive extreme pressures and temperatures using desiccation
- DSUP protects from radiation
 - Combined with desiccation allows for space survival



Future

- Tardigrades could travel through space
 - Possibly arrived from space
- Natural model for human space travel
 - DSUP protein mixed with human cell proved to protect human DNA from radiation
- Cryosleep by desiccation
- Applications of DSUP protein in chemotherapy



Supplementary slides: Dsup needs to be colocalized with DNA in order to protect

